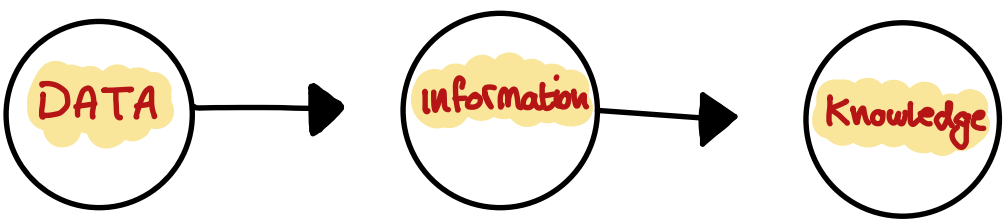
chapter one

- **Data**
  - A raw fact and can take the form of a number or statement
- **Information**
  - organized data, has meaning, prossesed data
- **when does data become information?**
  - when the data is processed
    - * Data + meaning = information
- **Knowledge**
  - Data and information that have been organized and processed
    - * information + application = Knowledge



- Systems
  - A collection of components that work together towords a common goal

    * Receive inputs $\xrightarrow{\text{transfer}}$ outputs

- Information systems
  - Combination of people, hardware, Software, Network and data

    * Providing info to managment to enable them to make dicisions

    * cims, library system, inventory system and bank

- CBIS - Computer Based information system
  - IS that uses computer technology
- BIS - Business information system
  - A group of interrelated components that work collectively to carry out Input, processing and output in order to covert data into information products

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- Input
  - The data flowing into the system from outside
- Processing
  - manipulating the input into a more useful form

- output
  - Results of the processing then exit the system

Storage
  - Holding information for use at a later date
    * • It is an output data
      • Input data
      Partly processed

- Feedback
  - when the outcomes has an influence on the input

The characteristics of valuable information

- Accurate
  Error-free
- Reliable
  can be depended on
- Simple
  غير معقد
- Timely
  found at the time needed
- Accessible
  can be used by authorized users
- Secure

Resources of a CBIS

A - People

- End user
  - → people who use IS
- IS specialist
  - → people who develope and operate IS

B - Hardware

- All physical devices and materials used in IS
  - → Input devices
  - → Output devices
  - → CPU - central processing unit
  - → memory

C - Software

- series of detailed instructions that control the operation of a computer system

systems software

1- operating software
- interact with hardware
- windows , linux , android

2- software development programs
- Allow users to develope their own software to carry out processing tasks

3- utility programs
- programs that monitor system performance or provid security controls

Application Software

1 - General - purpose applications

- programs that can be used to carry out a wide range of common tasks

2 - Application - specific software

- programs intended to serve a specific purpose

D - Data

- Records, files, reports, Documents, Database

E - Network

- Telecomuniction network, media
- A combination of devices connected to each other through links

- server
- Enduser
- Telecommunications
- middleware